

# AADK 13: Task 3 — 30-Day Compose Roadmap & Realization Plan.

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## 30-Day Compose Roadmap & Realization Plan

**Project Title:** Compose Heroes Hub

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### 1. 30-Day Project Vision

The **Compose Heroes Hub** is a Jetpack Compose application that allows users to explore a directory of superheroes with images, descriptions, and abilities. Users can **search heroes, filter favourites, switch themes, and view detailed hero profiles**.

The application will integrate **animations, accessibility support, responsive layouts, and automated testing** to demonstrate best practices in modern Android UI development using Jetpack Compose.

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### 2. Milestone Planning

The 30-day development plan is divided into **five milestones of six days each**, ensuring steady progress and manageable development phases.

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#### Milestone 1 — Project Setup & Core Architecture (Day 1–6)

**Goal:**

Establish the project foundation and create the core data structure.

**Tasks:**

- Set up Android Studio project using **Jetpack Compose**

- Configure **Material 3** theme system
- Implement **light and dark themes**
- Create **Superhero data model**
- Prepare sample dataset
- Configure **GitHub repository for version control**

**Expected Outcome:**

- Working Compose project with base theme and structure
- Data model ready for use in UI components

**Success Criteria:**

- Project builds successfully
  - Theme switching works correctly
  - Dataset loads without errors
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**Milestone 2 — Heroes List UI (Day 7–12)**

**Goal:**

Design and implement the primary list interface displaying superheroes.

**Tasks:**

- Create **Heroes List Screen**
- Implement **LazyColumn** for hero list
- Build reusable **HeroCard** composable
- Display hero image, name, and power summary
- Integrate image loading with **Coil**

**Expected Outcome:**

- Scrollable list of superheroes displayed smoothly
- Images load correctly with placeholders

**Success Criteria:**

- List displays correctly with at least **10–20 heroes**

- UI follows Material Design spacing and layout rules
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### **Milestone 3 — Search, Filter & Favourites (Day 13–18)**

#### **Goal:**

Introduce interactive features for filtering and user personalization.

#### **Tasks:**

- Add **search bar for hero name or powers**
- Implement **favourite toggle feature**
- Add **filter switch to display favourite heroes only**
- Manage UI state using **ViewModel and Kotlin Flow**
- Ensure UI updates dynamically based on user actions

#### **Expected Outcome:**

- Users can easily find heroes using search
- Favourite heroes persist within app state

#### **Success Criteria:**

- Search results update in real time
  - Favourite toggles update UI immediately
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### **Milestone 4 — Detail Screen & Animations (Day 19–24)**

#### **Goal:**

Enhance the application with navigation and smooth UI interactions.

#### **Tasks:**

- Implement **Hero Detail Screen**
- Add **navigation between list and detail screens**
- Implement **animations for list items**
- Add **favourite animation feedback**
- Improve layout responsiveness

**Expected Outcome:**

- Users can tap a hero to view detailed information
- Smooth animations improve UI experience

**Success Criteria:**

- Navigation works reliably
  - Animations run smoothly without lag
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**Milestone 5 — Accessibility, Testing & Optimization (Day 25–30)****Goal:**

Ensure the application meets accessibility standards and performance goals.

**Tasks:**

- Conduct **accessibility audit**
- Add **content descriptions for images**
- Ensure **touch targets  $\geq 48dp$**
- Implement **Compose UI automated tests**
- Optimize image loading and memory usage
- Prepare **final project presentation**

**Expected Outcome:**

- Accessible and testable Compose application
- Stable performance even with larger datasets

**Success Criteria:**

- Accessibility checklist passes
  - Automated tests run successfully
  - App performs smoothly on mid-range devices
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### 3. Resource & Tooling Plan

#### Development Tools

- **Jetpack Compose** – UI framework
  - **Android Studio Hedgehog / latest version**
  - **Coil** – Image loading and caching
  - **ViewModel + Kotlin Flow** – State management
  - **Accompanist Placeholder** – Loading placeholders
  - **GitHub** – Version control and project management
  - **CI pipeline** – Automated testing using GitHub Actions
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#### Learning Resources

The following resources will be used throughout the project:

1. Official **Android Developers Jetpack Compose documentation**
2. **30 Days of Compose codelab**
3. Android Developers **YouTube tutorials**
4. Medium articles on **Compose performance optimization**
5. GitHub open-source Compose projects for reference

These resources will support continuous learning and troubleshooting during development.

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### 4. Risk & Mitigation Table

| Risk                                     | Impact                                | Mitigation Strategy                                             |
|------------------------------------------|---------------------------------------|-----------------------------------------------------------------|
| Large dataset slows UI performance       | List scrolling may lag                | Use LazyColumn, optimize images, implement pagination if needed |
| Animations causing jank on older devices | Poor user experience                  | Use lightweight animations and test on mid-range devices        |
| Accessibility issues discovered late     | Difficult to fix after implementation | Conduct weekly accessibility checks                             |

| Risk                  | Impact                      | Mitigation Strategy                      |
|-----------------------|-----------------------------|------------------------------------------|
| State management bugs | UI may not update correctly | Use ViewModel and structured state flows |

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## 5. Reflection & Commitment

This 30-day project is exciting because it allows me to apply multiple Jetpack Compose concepts into a single cohesive application. Building the **Compose Heroes Hub** will help strengthen my understanding of modern Android UI architecture, state management, and responsive design.

I expect to learn how to integrate **animations, accessibility practices, and automated testing** into real-world applications. Progress will be measured through milestone completion, successful test execution, and weekly reviews of implemented features.

By following this structured roadmap, I aim to build a **fully functional, accessible, and well-tested Compose application within 30 days**.